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# Applied Data Science Project

L12 – Colaboratory

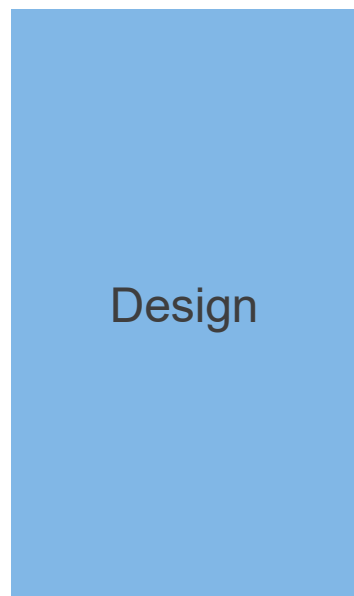


**Politecnico  
di Torino**

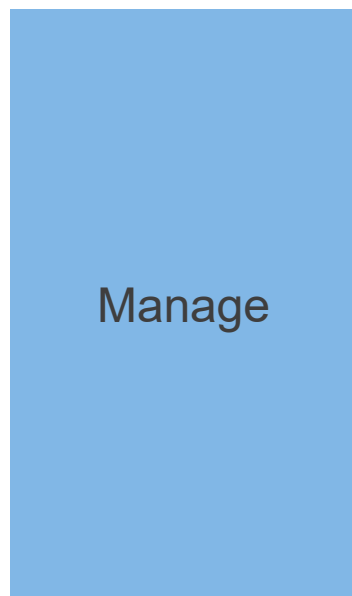


**e l l i s**  
European Laboratory for Learning and Intelligent Systems

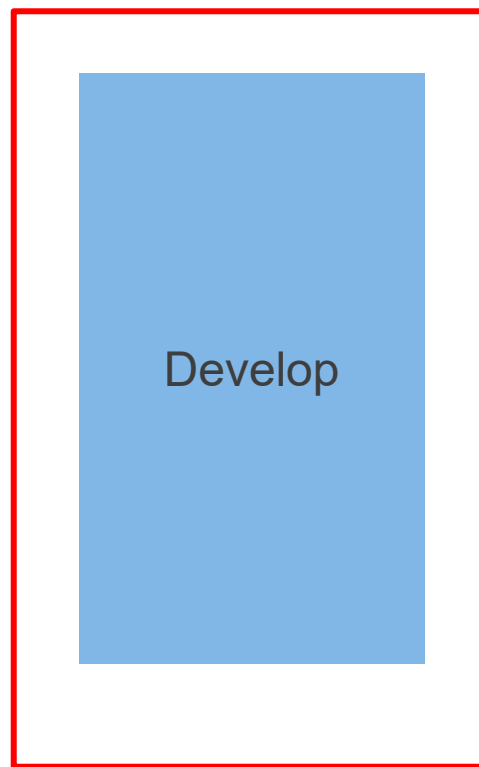
# Pillars



Design



Manage



Develop



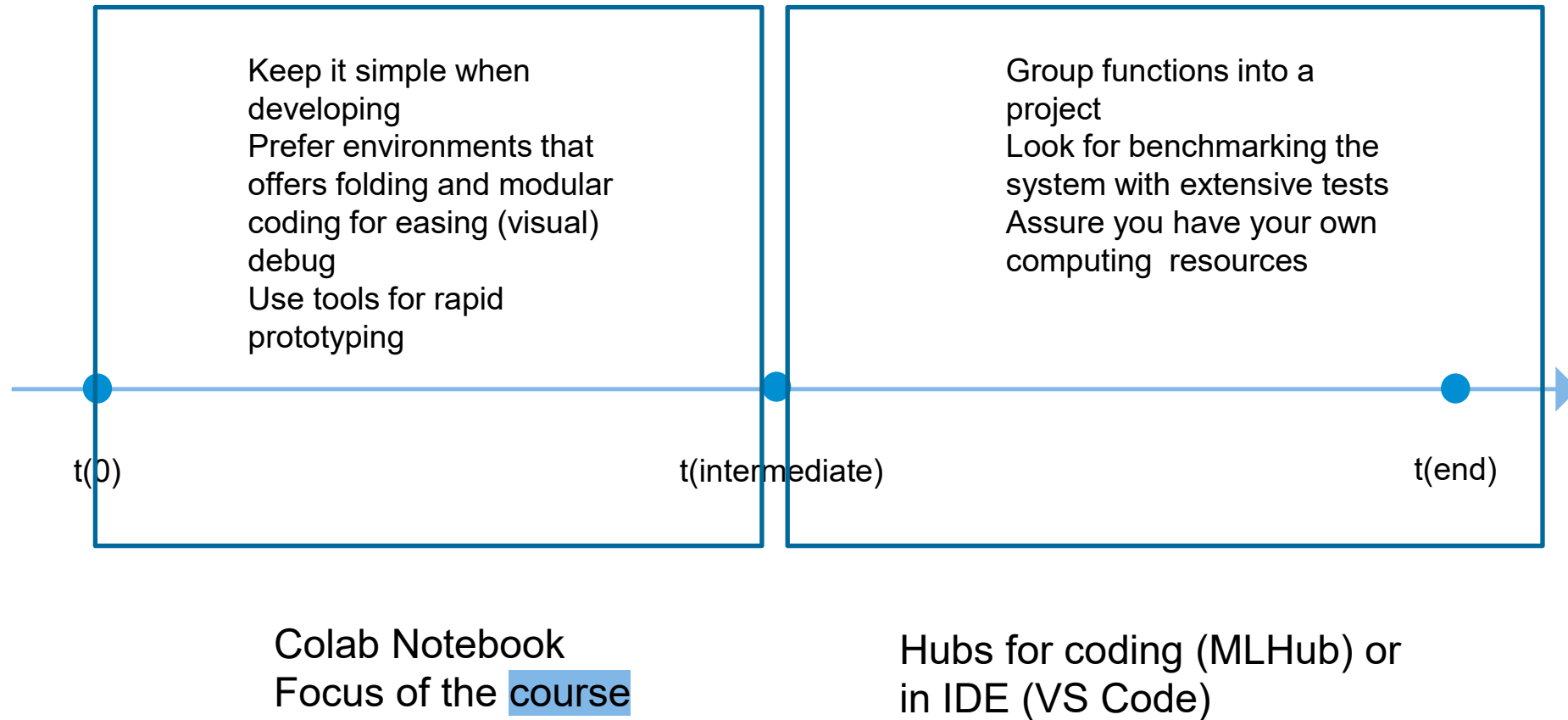
Communicate

# Development

- “Divide and rule”
- Tasks are mapped into modules
- One module has one lead developer and, eventually, contributors
- Choose the programming language according to:
  - ecosystem of software modules that you can utilize
  - easiness of model integration
  - familiarity. Do not be afraid to switch to another (similar programming language) since they share most of the features and development patterns

Python is considered the default language for developing machine intelligence nowadays

# Development timeline



# Colaboratory for rapid prototyping



<https://colab.research.google.com>

An application where to develop, share, and also test on dedicated hardware (GPU to speed up the computing)



# Overview

The screenshot shows the Google Colaboratory web interface. At the top, the title bar says "Welcome To Colaboratory" with a menu: File, Edit, View, Insert, Runtime, Tools, Help. On the left is a sidebar with a "Table of contents" panel containing links to "Getting started", "Data science", "Machine learning", "More Resources", "Machine Learning Examples", and a "Section" button. The main workspace has a toolbar with "+ Code" and "+ Text" buttons, a "Copy to Drive" button, and a "Share" button with a settings gear. Annotations with arrows point to these elements: "add a block for writing explanations" points to the "+ Text" button; "add a block for code writing" points to the "+ Code" button; "share with collaborators" points to the "Share" button. The main content area displays a notebook titled "What is Colaboratory?". It contains a paragraph about Colab, a bulleted list of features (Zero configuration required, Free access to GPUs, Easy sharing), and a section titled "Getting started" which explains the interactive environment and includes a code cell with a Python script to calculate seconds in a day.

add a block for writing explanations

add a block for code writing

share with collaborators

Table of contents of the notebook

Notebook containing both code blocks and explanations in a narrative fashion

**Table of contents**

- Getting started
- Data science
- Machine learning
- More Resources
- Machine Learning Examples
- Section

**What is Colaboratory?**

Colaboratory, or "Colab" for short, allows you to write and execute Python in your browser, with

- Zero configuration required
- Free access to GPUs
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch [Introduction to Colab](#) to learn more, or just get started below!

### Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
[ ] seconds_in_a_day = 24 * 60 * 60
    seconds_in_a_day

86400
```

To execute the code in the above cell, select it with a click and then either press the play button to the left of the code, or use the keyboard shortcut "Command/Ctrl+Enter". To edit code, just click the cell and start editing.

Variables that you define in one cell can later be used in other cells:

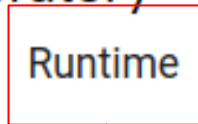
```
[ ] seconds_in_a_week = 7 * seconds_in_a_day
    seconds_in_a_week
```

# Menu bar



Welcome To Colaboratory

File Edit View Insert Runtime Tools Help



define the runtime configurations, among those the hardware acceleration if needed

## Notebook settings

Hardware accelerator

GPU 

To get the most out of Colab, avoid using a GPU unless you need one. [Learn more](#)

☐ Omit code cell output when saving this notebook

Cancel

Save

# Share

 **Simple Sentiment Analysis.ipynb**   Comment  **Share** 

File Edit View Insert Runtime Tools Help

 **Share with people and groups** 

Add people and groups

 **Giuseppe Rizzo (you)** Owner

[Send feedback to Google](#)

Done

 **Get link**

**Restricted** Only people added can open with this link

[Change to anyone with the link](#)

[Copy link](#)

# Connect your Google Drive

Enable authorization to import data from Google Drive

```
1 from google.colab import drive
2 drive.mount('/content/drive/')
```

List the files in your drive

```
1 !ls "/content/drive/My Drive/"
```

# Running with Google Colab

Upload YOUR\_PYTHON\_FILE.py to Google Drive & Run with Google Colab

```
1 !python3 "/content/drive/My Drive/Colab  
  Notebooks/YOUR_PYTHON_FILE.py"
```

Run with Google Colab to Download YOUR\_PYTHON\_FILE.py from Google Drive

```
1 from google.colab import files  
2 files.download('/content/drive/My Drive/Colab  
  Notebooks/YOUR_PYTHON_FILE.py')
```

# Bash commands

Bash commands are executed with the environment “!”

Download an external file

```
1 !wget
  http://ai.stanford.edu/~amaas/data/sentiment/aclImdb\_v1.tar.gz -P "/content/drive/My Drive/Colab Notebooks"
```

Clone a repository

```
1 !git clone https://github.com/pytorch/examples.git
```

# Colab == virtual environment

The environment can be customized with the addition of python packages

## Install

```
1 !pip install -q datasets
```

## Show a version

```
1 !pip show datasets
```





# Thank you for your attention.

Questions?



# CONTACTS

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